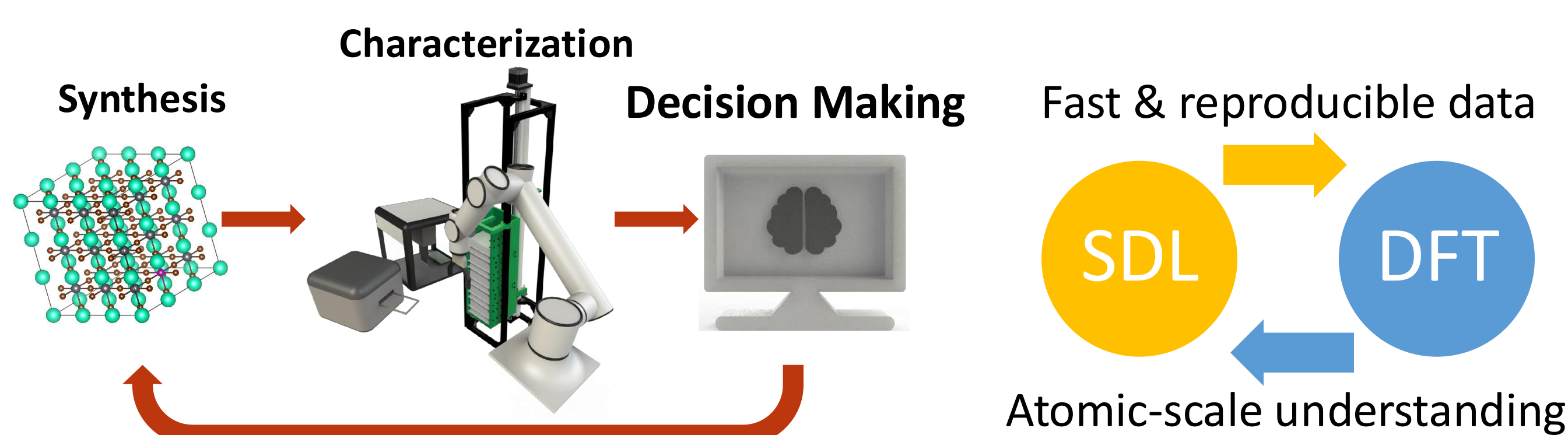


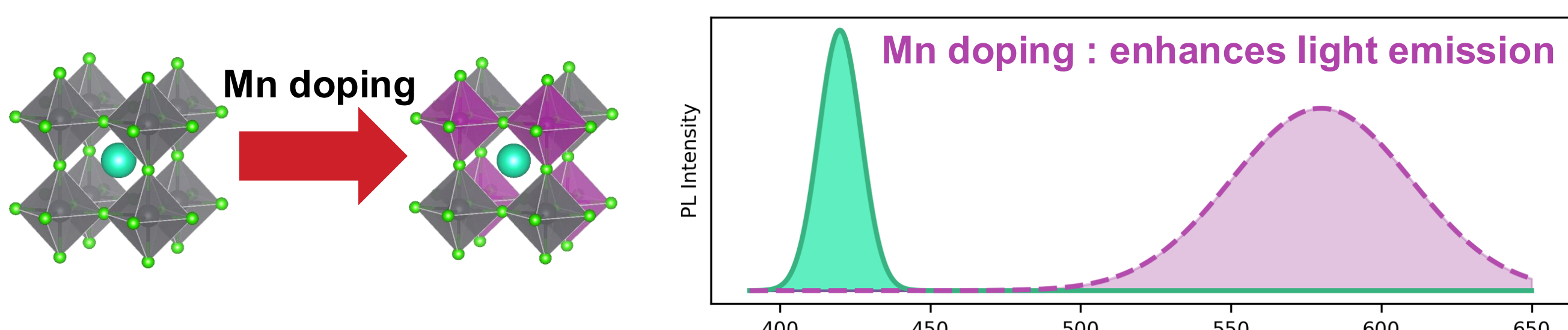
Self Driving Labs (SDL)



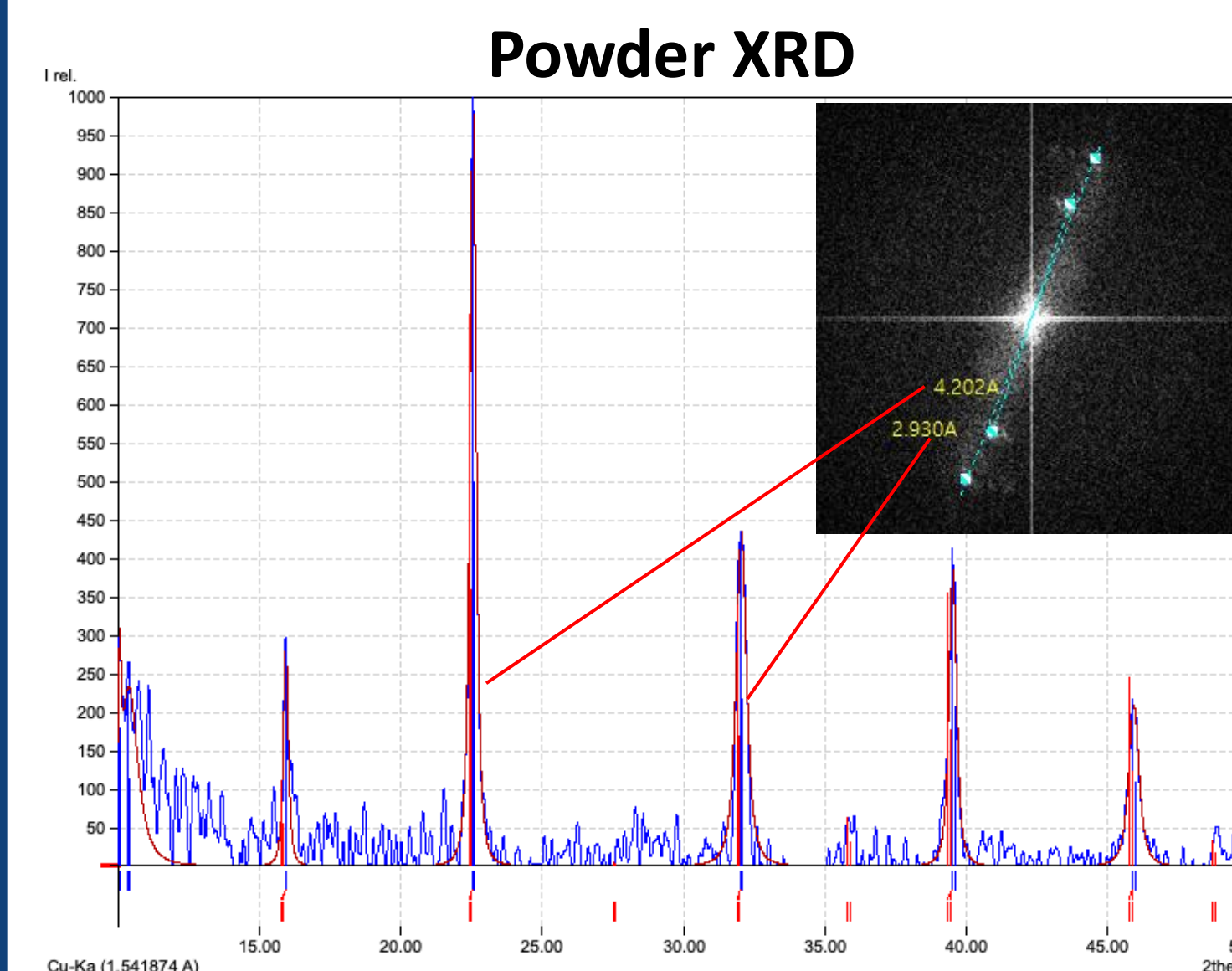
- While SDL can optimize optical properties, it lacks ab initio mechanistic insight.
- Density Functional Theory (DFT) calculation can provide microscopic understandings.

Mn doped CsPbCl_3

- Mn^{2+} : **dual emission**, easy to integrate into NCs.
- Utilize SDL to investigate chemical space quickly and maximize photoluminescence quantum yield.



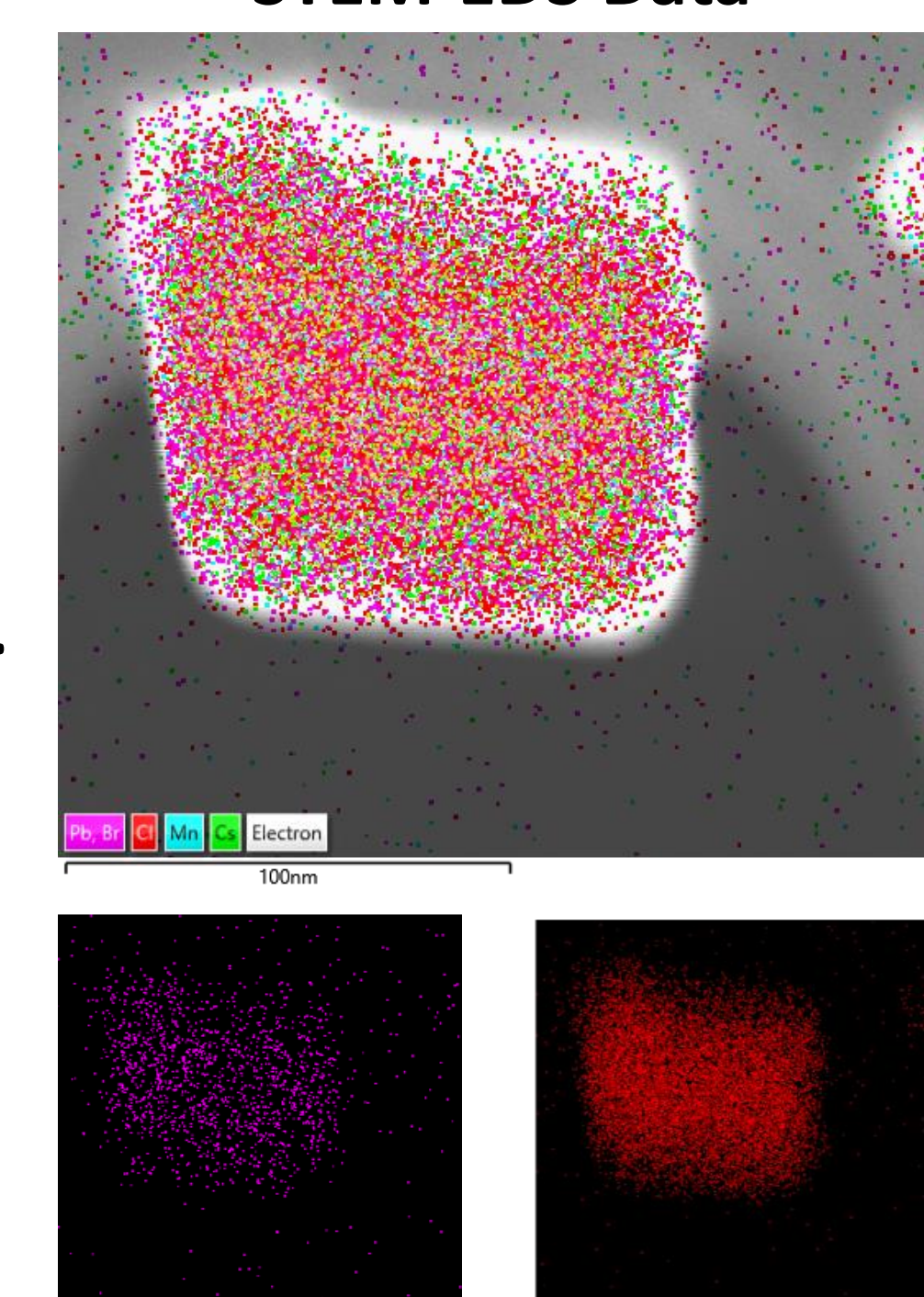
Characterization



ICP-MS Data	
	Atomic ratio (%)
Mn	3.4
Cs	43.0
Pb	53.6
Doped %	5.95

	Atomic ratio (%)
Mn	6.0
Cs	47.8
Pb	46.2
Doped %	11.49

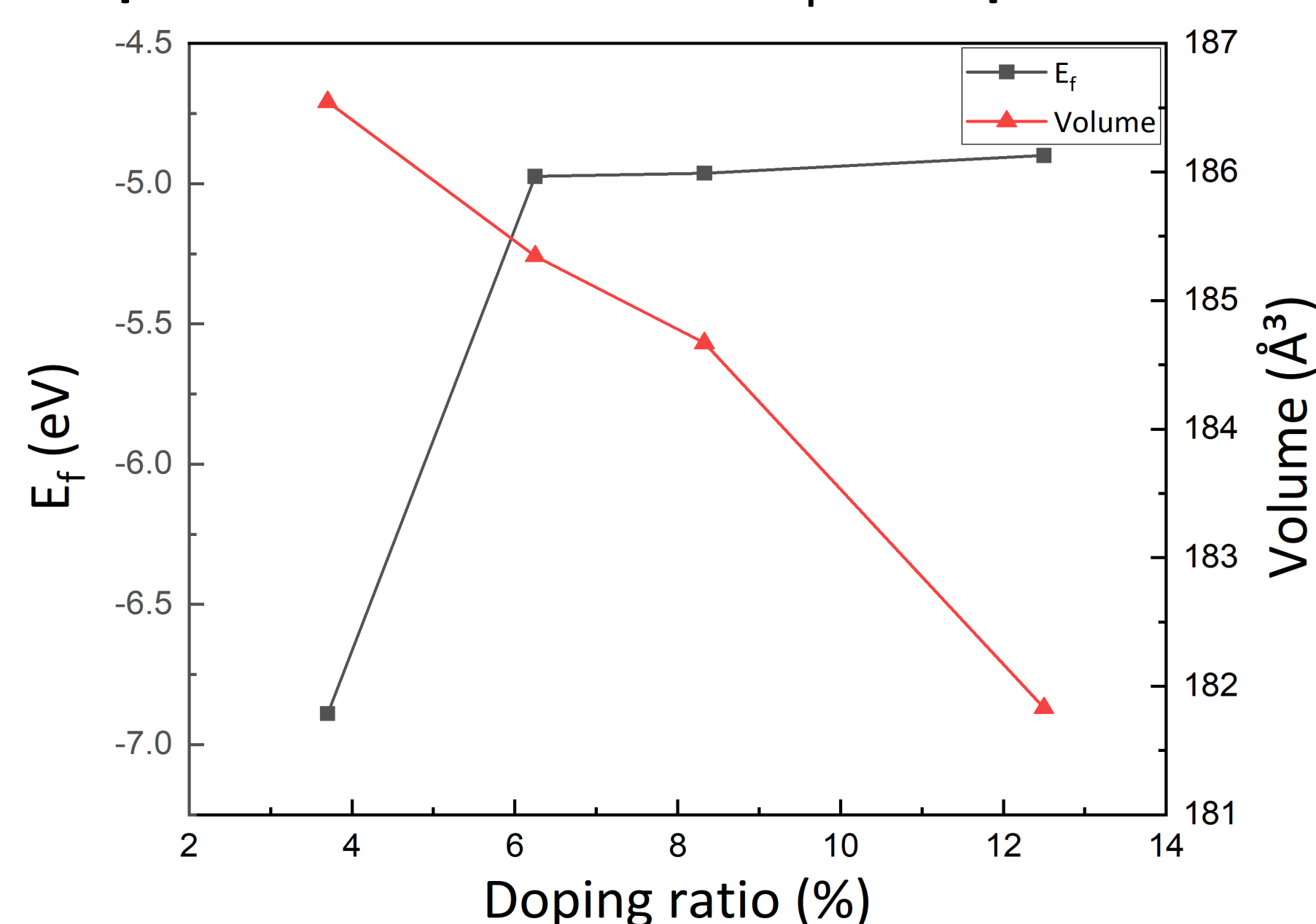
STEM-EDS Data



- XRD and HRTEM FFT images validate that nanocrystal's main structure is CsPbCl_3 .
- ICP-MS and STEM-EDS revealed the **atomic ratios** of nanocrystals synthesized via SDL.
- This characterization provides the quantitative parameters required to translate nanocrystals into computational model.

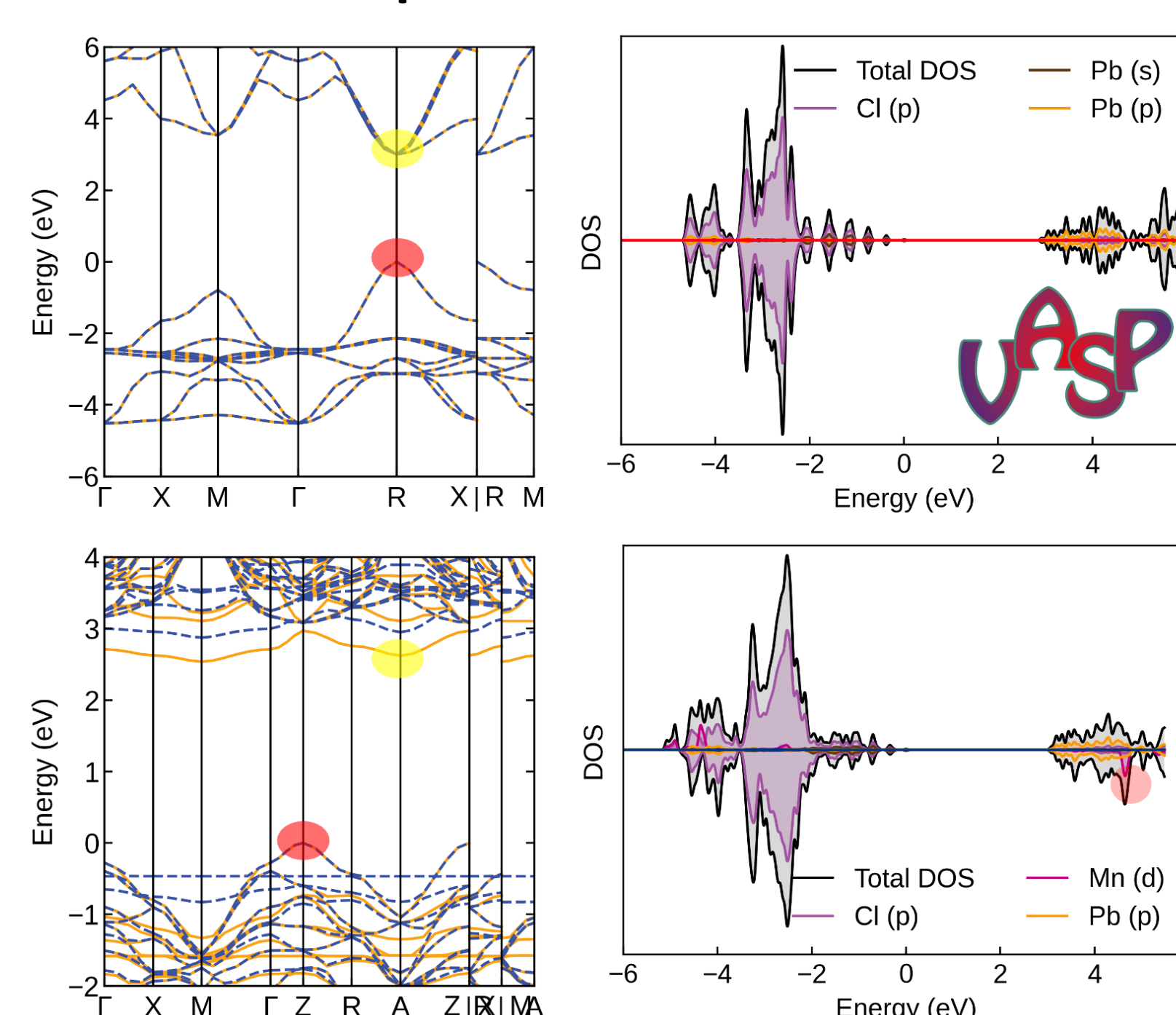
Computation Modeling

Dependence of Volume and E_f on Dopant Ratio



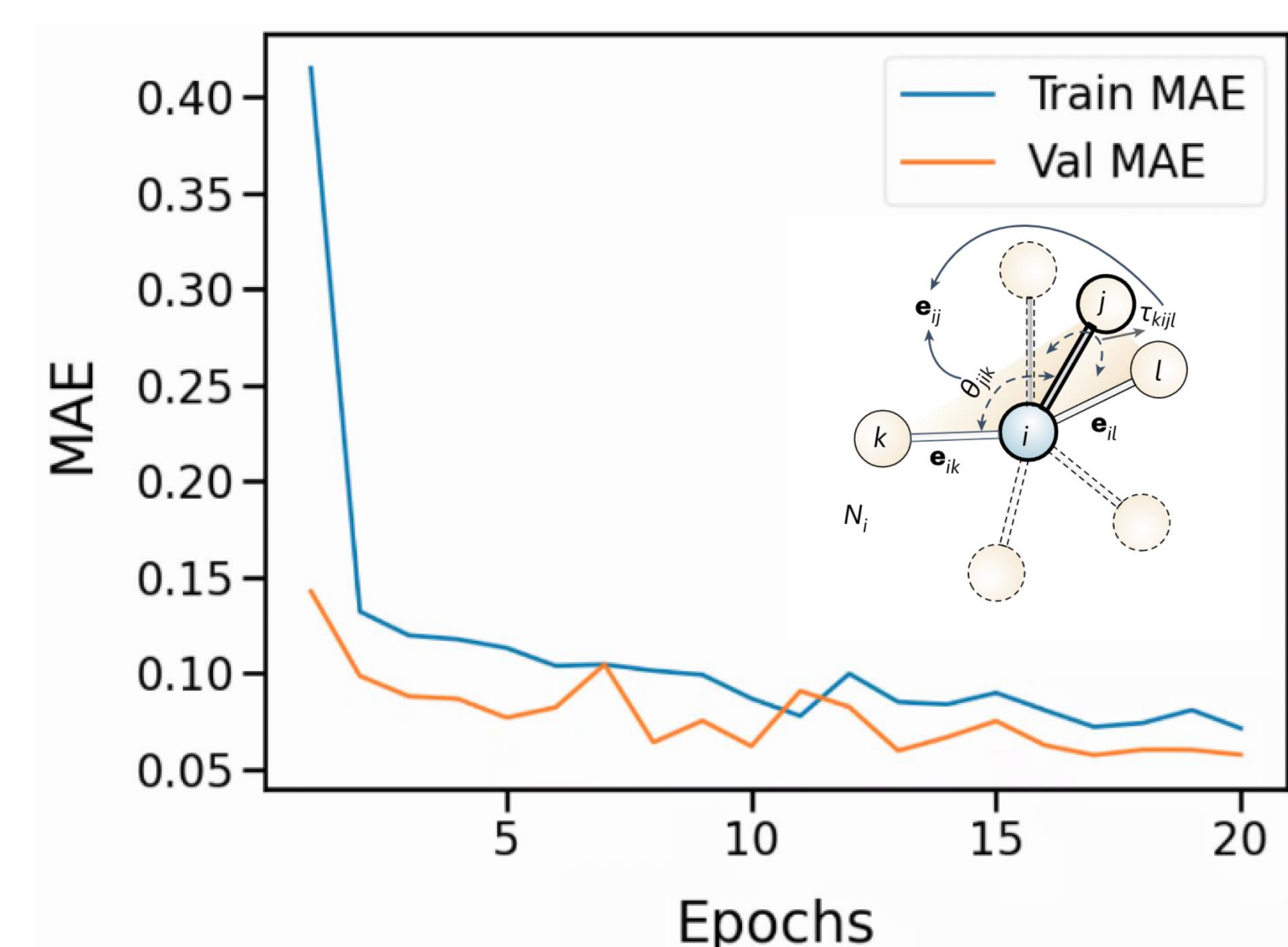
- Lattice contraction arises from two competing distortions: the MnCl_6 octahedron shrinks, PbCl_6 octahedron slightly expand.
- Controlled Mn doping produces nanocrystals that retain their structural stability.

Effect of dopant on Electronic Structure



- While pristine CsPbCl_3 exhibits a direct bandgap, Mn-doped CsPbCl_3 transitions to an indirect bandgap, due to **Mn-d orbitals that distort** the band structure and the density of states.

Performance of fine-tuned M3GNet Model



- Fine tuned GNN models based on M3GNet using DFT calculated structure data.
- The incorporation of DFT data allows the model to reliably **predict formation-energy** in perovskite nanocrystal system.

Conclusion

- Our results confirm that SDL-synthesized nanocrystals possess thermodynamic and structural stability, **validating the reliability** of SDL-based synthesis.
- This workflow demonstrates how automated experimentation and computational modeling can be combined to accelerate microscopic understanding and macroscopic optimization.
- Incorporating DFT as a **high-level decision layer** within the SDL pipeline is anticipated to greatly enhance and accelerate the identification of optimal synthesis pathways.
- This research was supported by the Independent Research Program of the SNU College and by North Carolina State University.